Review of P13 – MS1

Project: Padel Prediction: Pre match win probabilities for the Premier Padel / FIP tour

1 - Summary

The project wants to predict the win probability for men's and women's Premier Padel tour and selected FIP Tour matches. It uses historical match data, player ratings, rankings, rest and partnership information to make daily predictions, while comparing its performance against the existing Elo-based probability baseline provided by padelapi.org. The proposal also includes a complete MLOps Workflow with automated feature generation, model training, inference, monitoring, model registration and cloud-based serving.

2 - Strengths

One strong point is that the success criterion is clearly measurable and connected to a meaningful baseline. Instead of only targeting a generic accuracy value, the proposed model must achieve a lower out-of-time log loss than the provider's existing Elo model. This makes it clear what additional value the new features are expected to provide.

Another strength is the handling of time-based data and leakage. Features such as Elo, form, rest, partnership history and rankings are calculated only with information available before each match. The proposal also uses an out-of-time split at tournament boundaries and a unit test checking that changing a match result does not change the feature of the same match. This helps to reduce the risk of data leakage.

The system design is well structured. The project separates features, training, and inference pipelines, together with Hopsworks, Weights & Biases, GitHub Actions and Cloud Run creates a clear end-to-end MLOps workflow. It is also good that tools like Granfana, Evidently and Terraform are only planned as optional stretch goals.

3 - Concerns or gaps

One small point that could be clarified further is how player-level values are combined into team-level features. For example, "form_diff", "rest_days_diff" and some partnership-related information are listed, but the exact calculations on the two individual players within a team is not fully described. This is not a major issue at the proposal stage, but documenting it would make the feature pipeline easier to understand and reproduce.

4 - One concrete suggestion

Before MS2, I would suggest adding a short description of how these team-level features are calculated. For example the documentation could explain how the two players values are combined and how many previous matches are used for recent form. This would make the feature engineering more clear without adding much extra work.